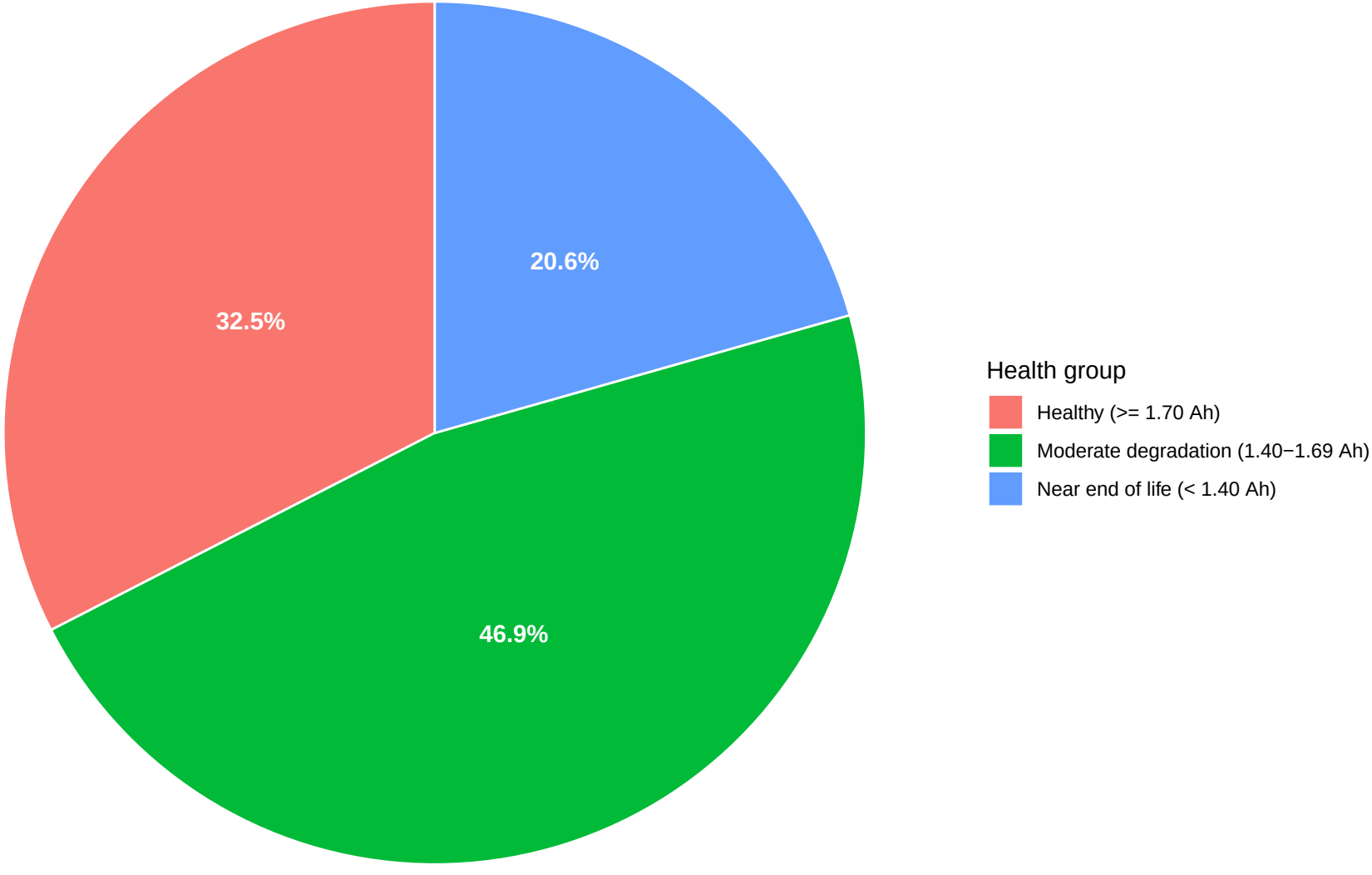
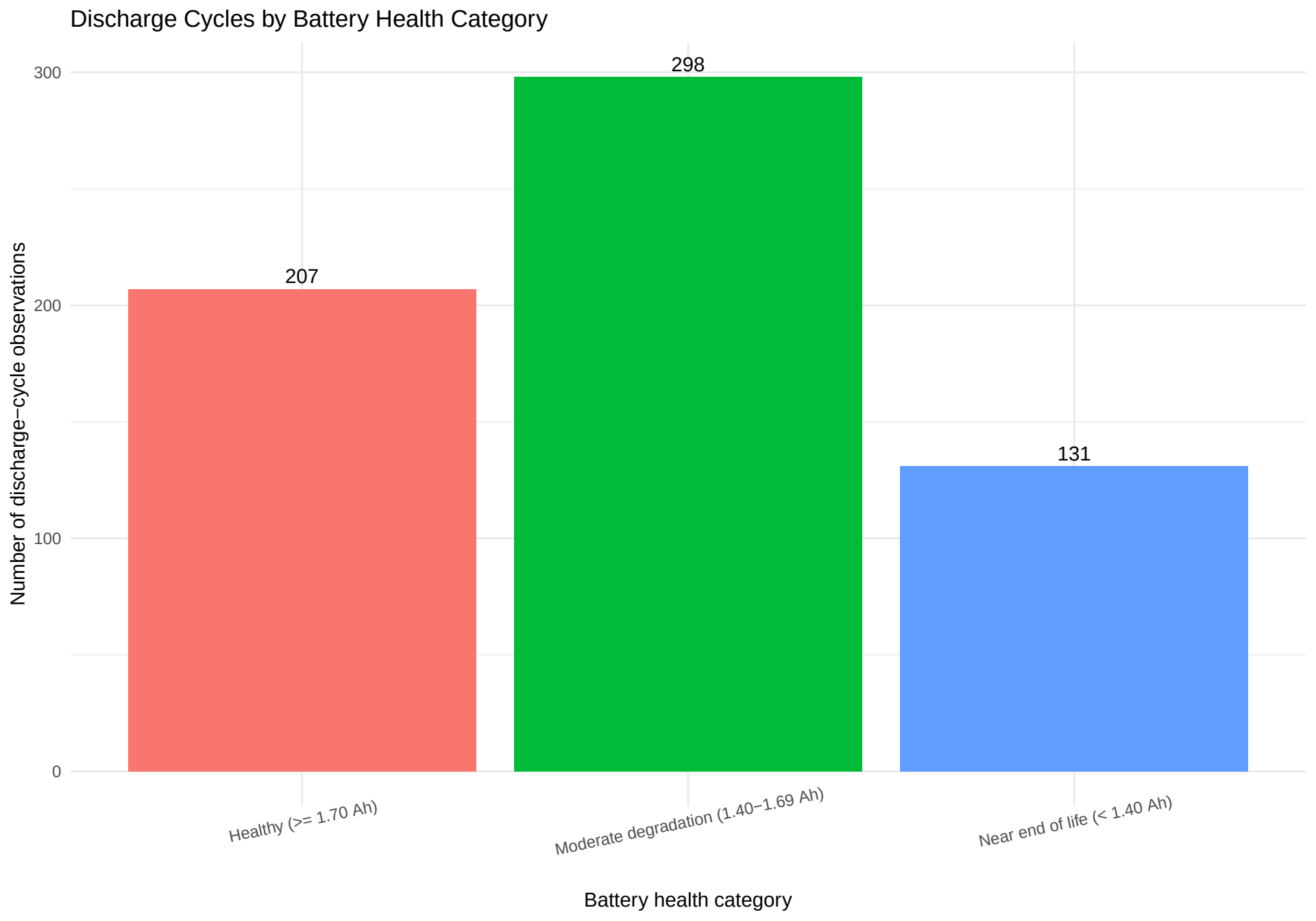
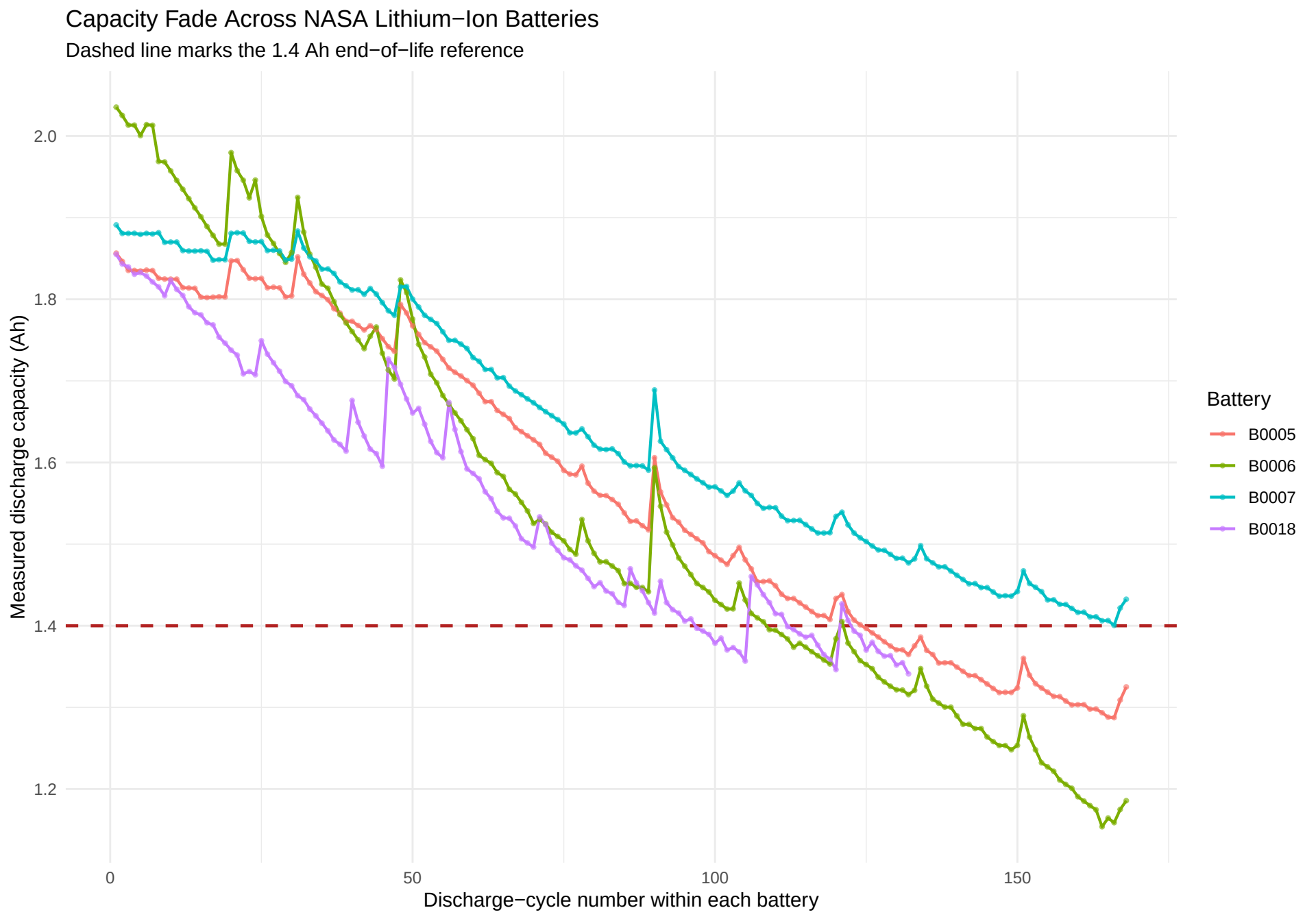
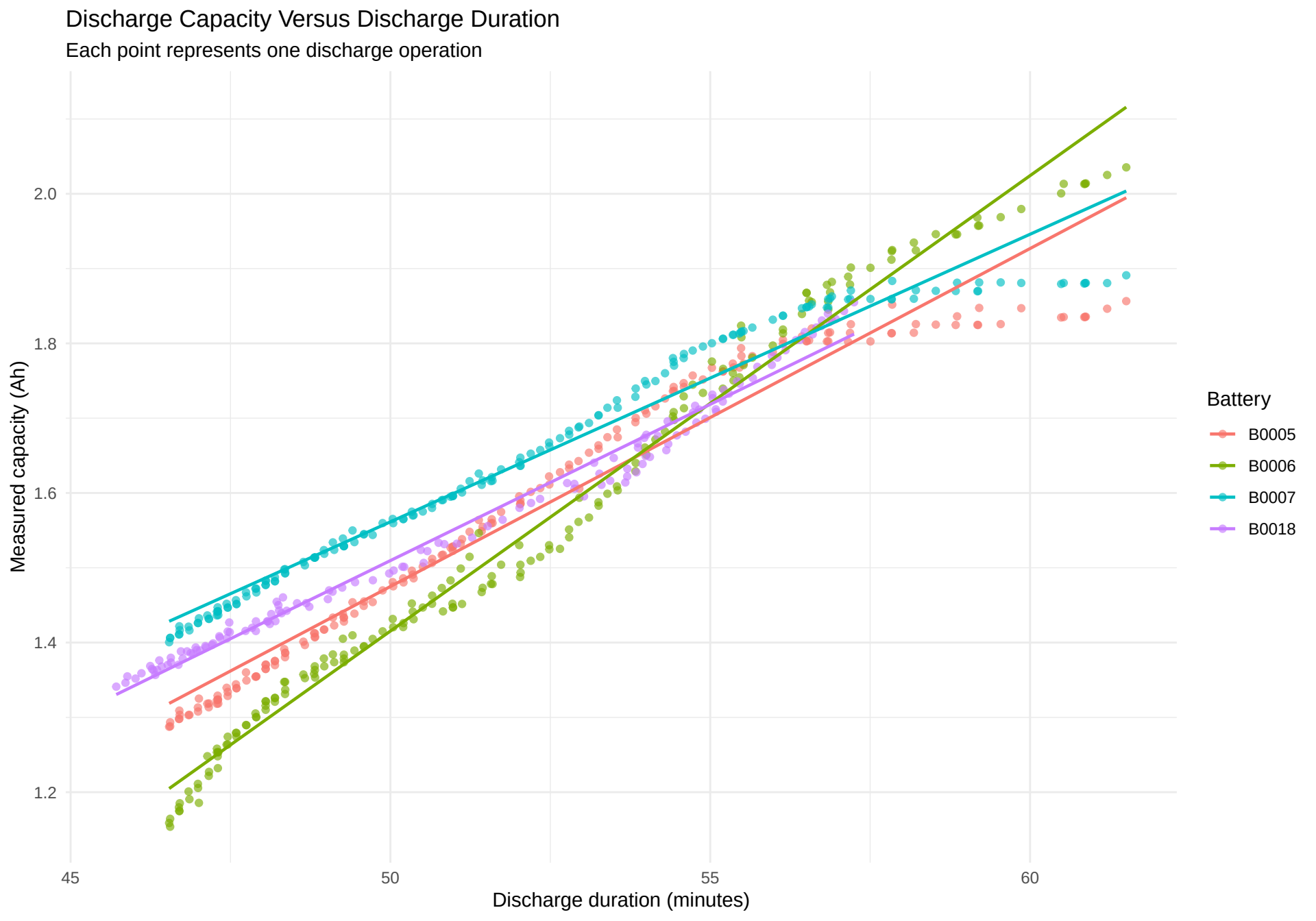
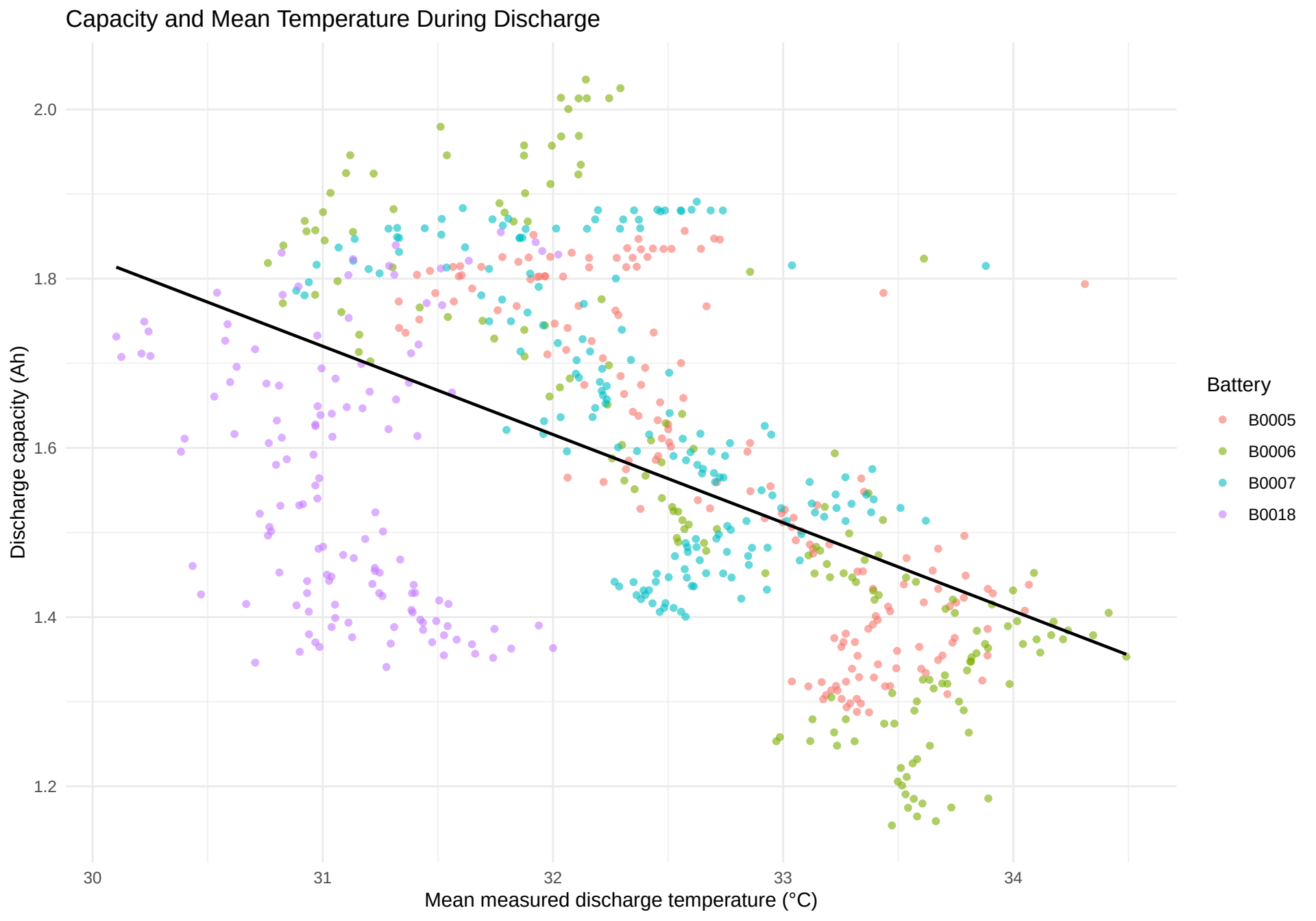
Distribution of Discharge Cycles by Battery Health

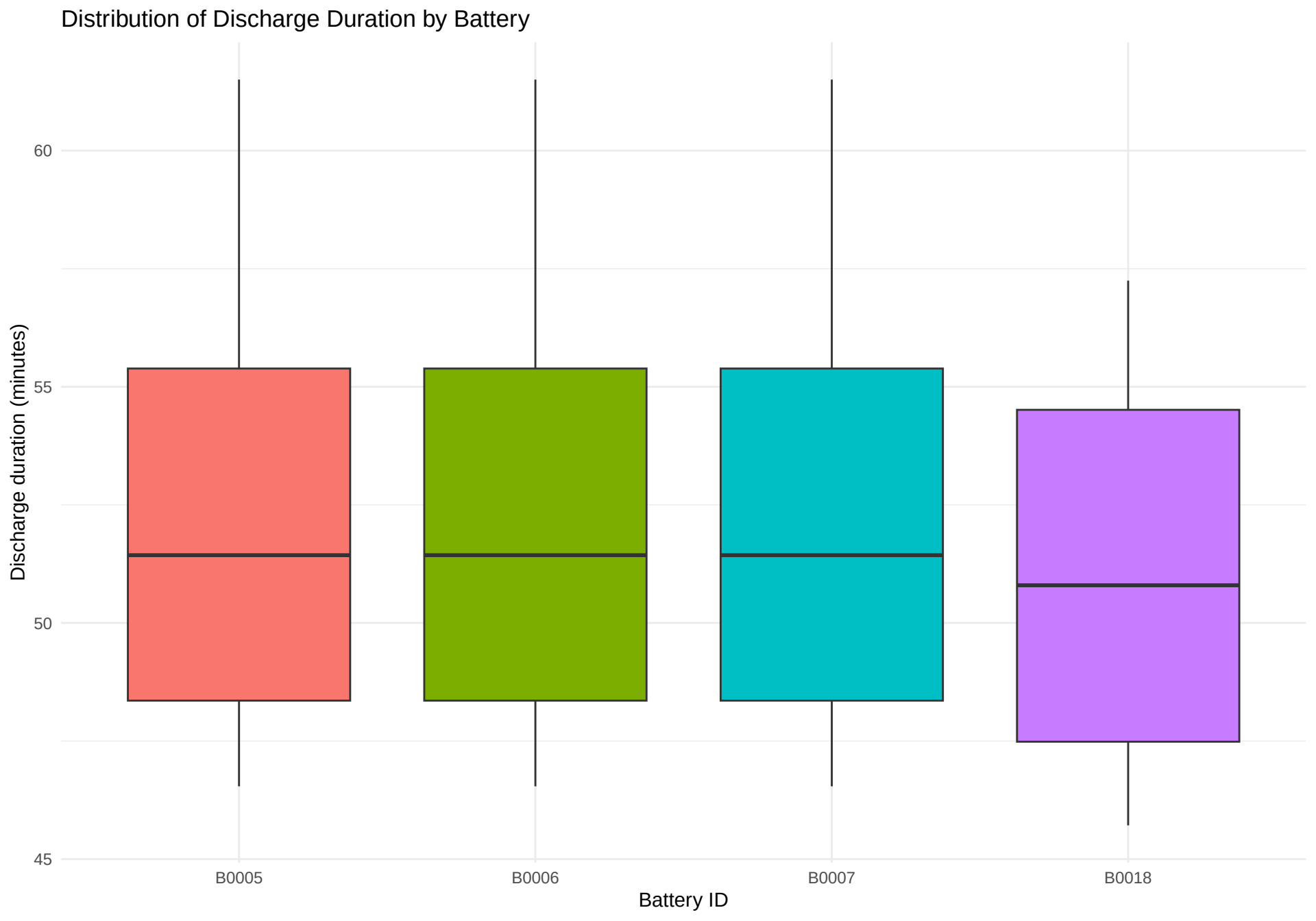


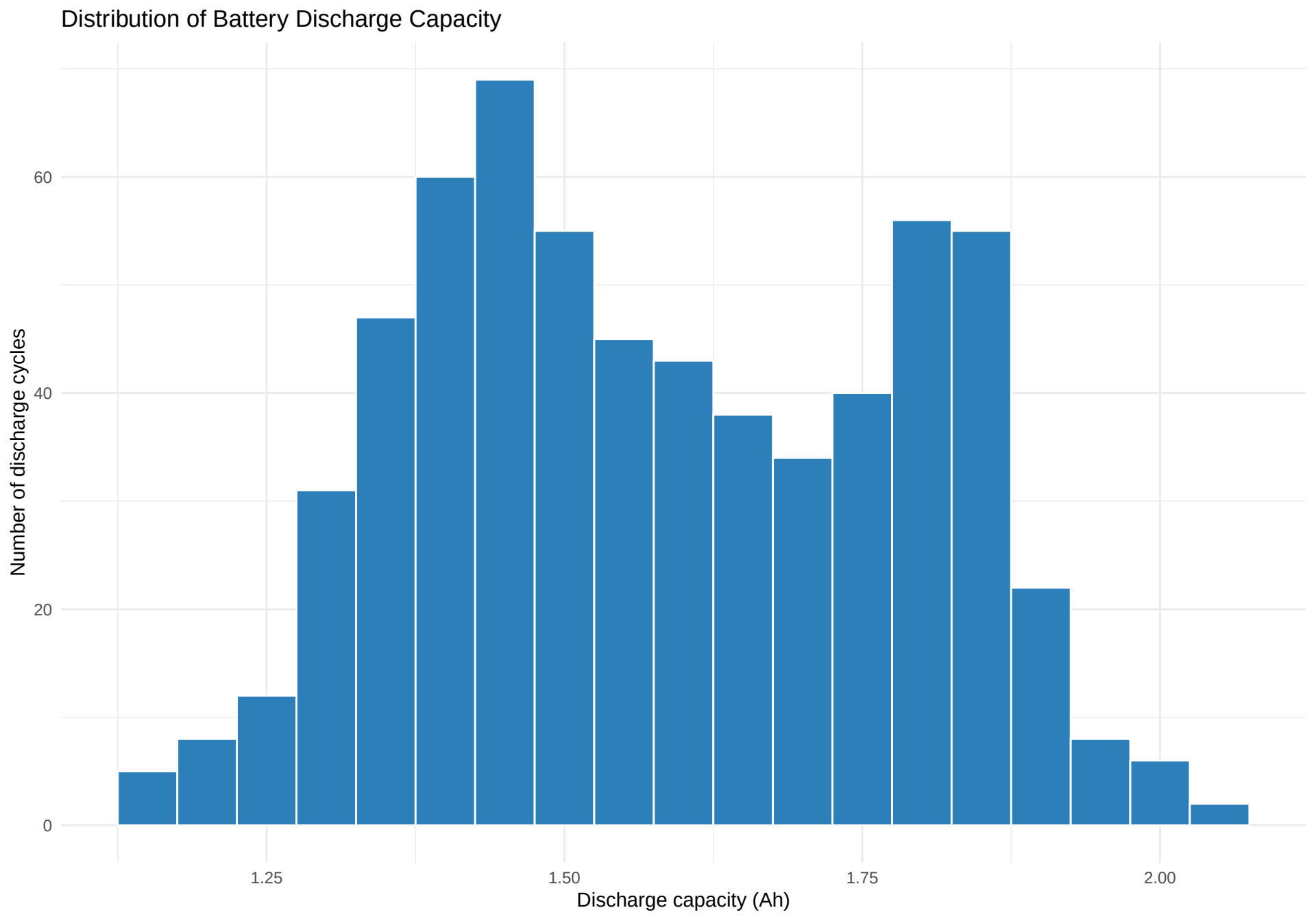

Discharge Cycles by Battery Health Category
300
200
100
0
Number of discharge−cycle observations
207
298
131
Healthy (>= 1.70 Ah)
Moderate degradation (1.40–1.69 Ah)
Near end of life (< 1.40 Ah)
Battery health category

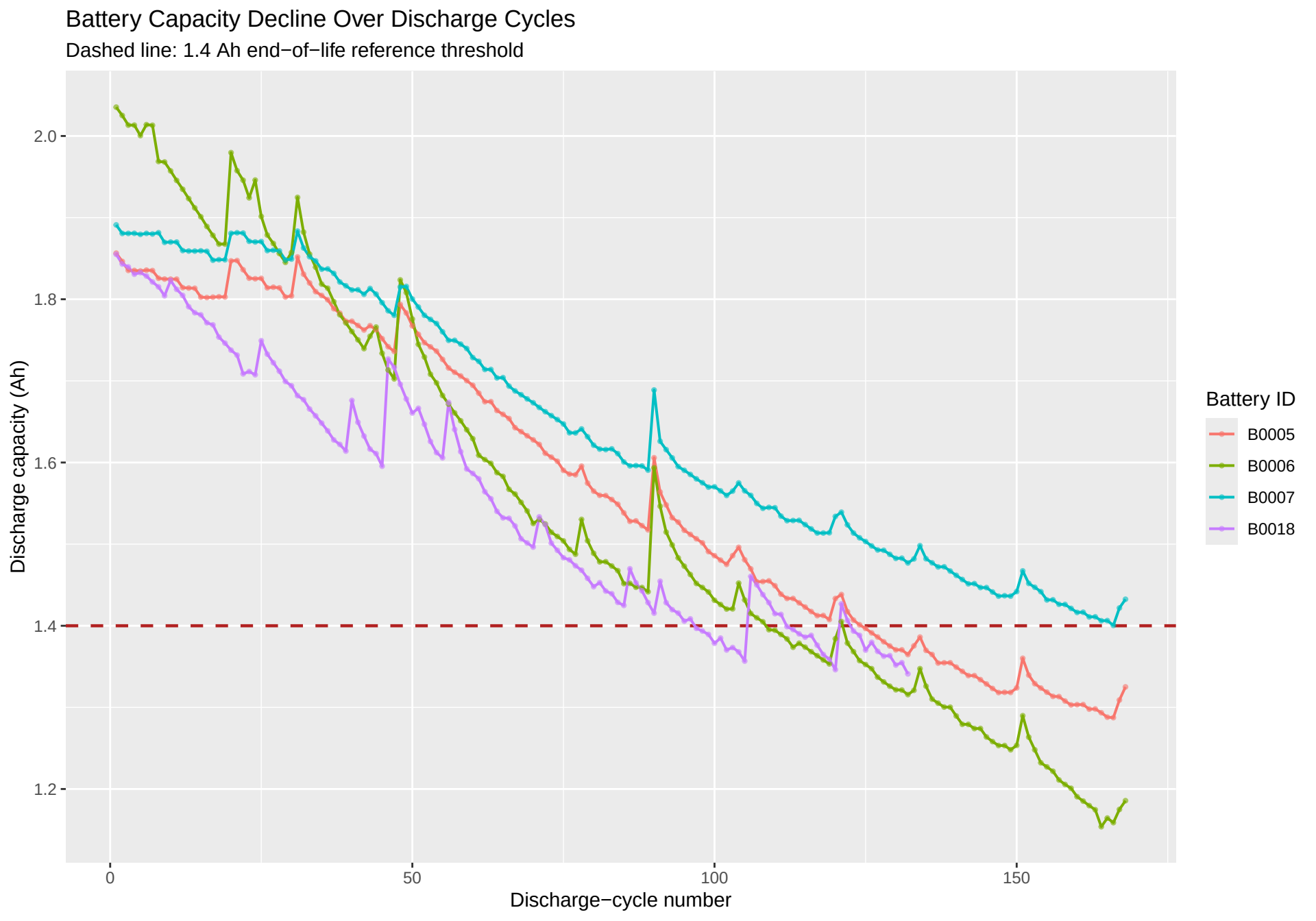

Capacity Fade Across NASA Lithium−Ion Batteries
Dashed line marks the 1.4 Ah end−of−life reference
Measured discharge capacity (Ah)
2.0
1.8
1.6
1.4
1.2
0
50
100
150
Discharge−cycle number within each battery
Battery
B0005
B0006
B0007
B0018

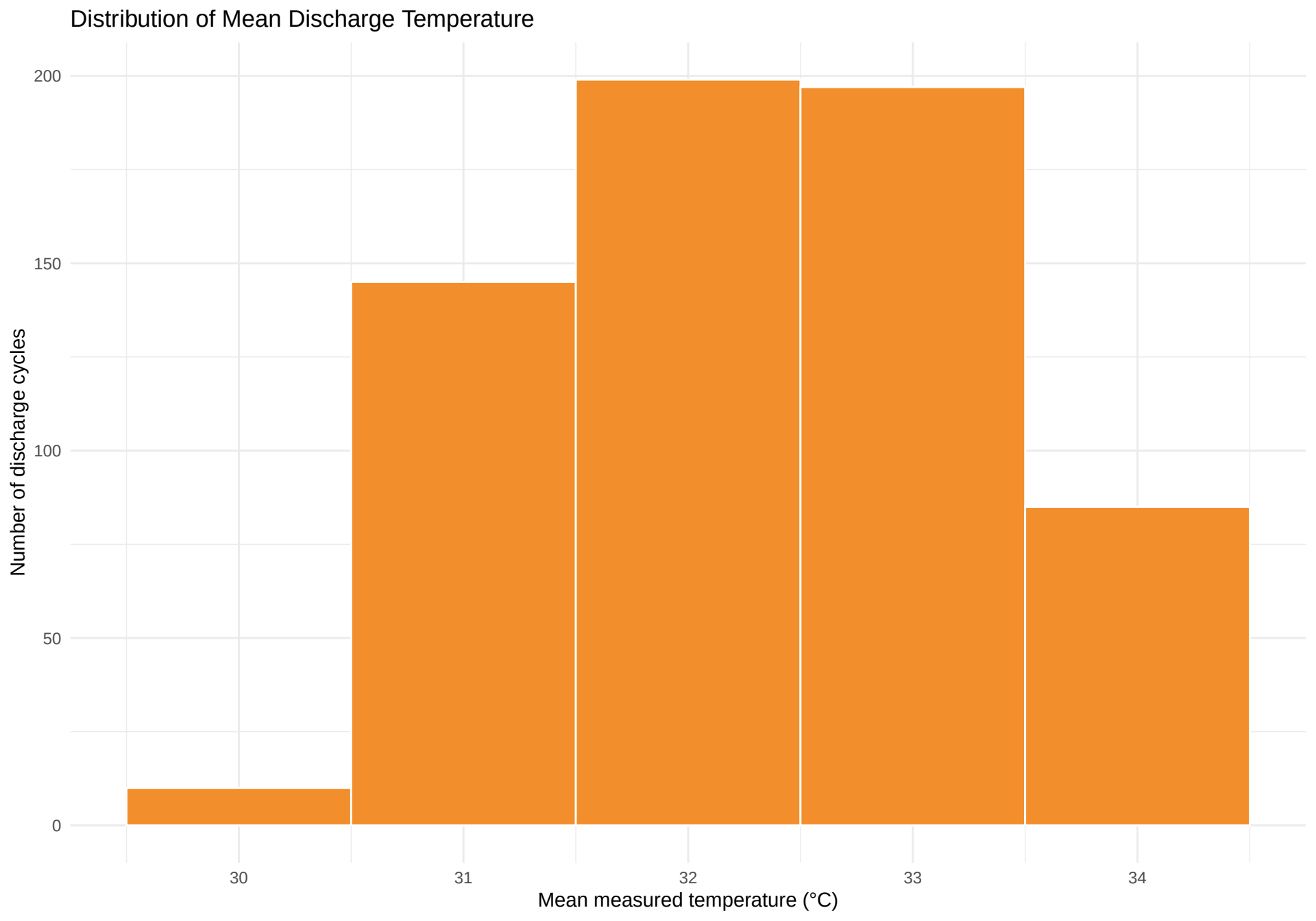

Discharge Capacity Versus Discharge Duration
Each point represents one discharge operation
Measured capacity (Ah)
2.0
1.8
1.6
1.4
1.2
45
50
55
60
Discharge duration (minutes)
Battery
B0005
B0006
B0007
B0018

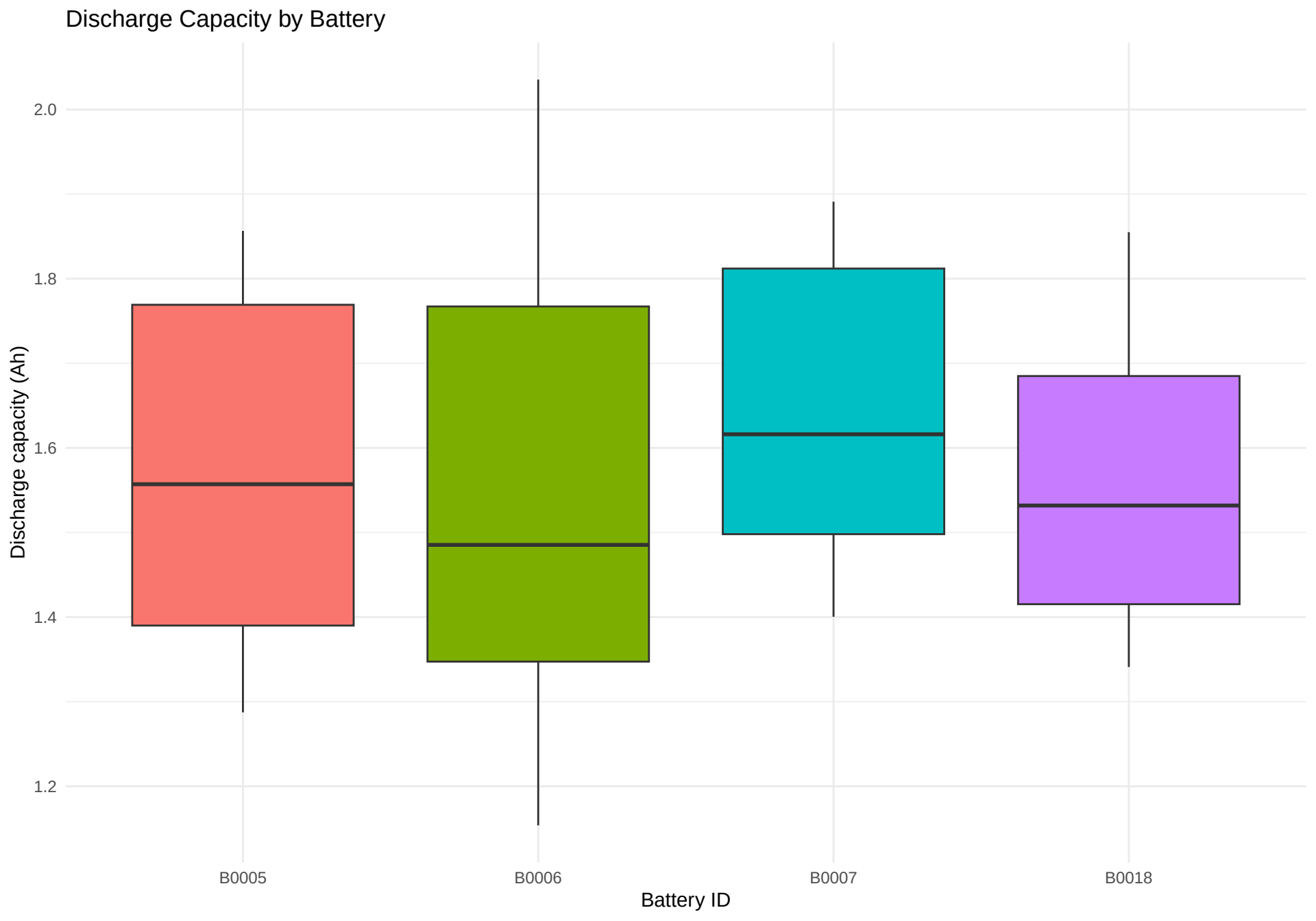

Capacity and Mean Temperature During Discharge
Discharge capacity (Ah)
2.0
1.8
1.6
1.4
1.2
30
31
32
33
34
Mean measured discharge temperature (°C)
Battery
B0005
B0006
B0007
B0018

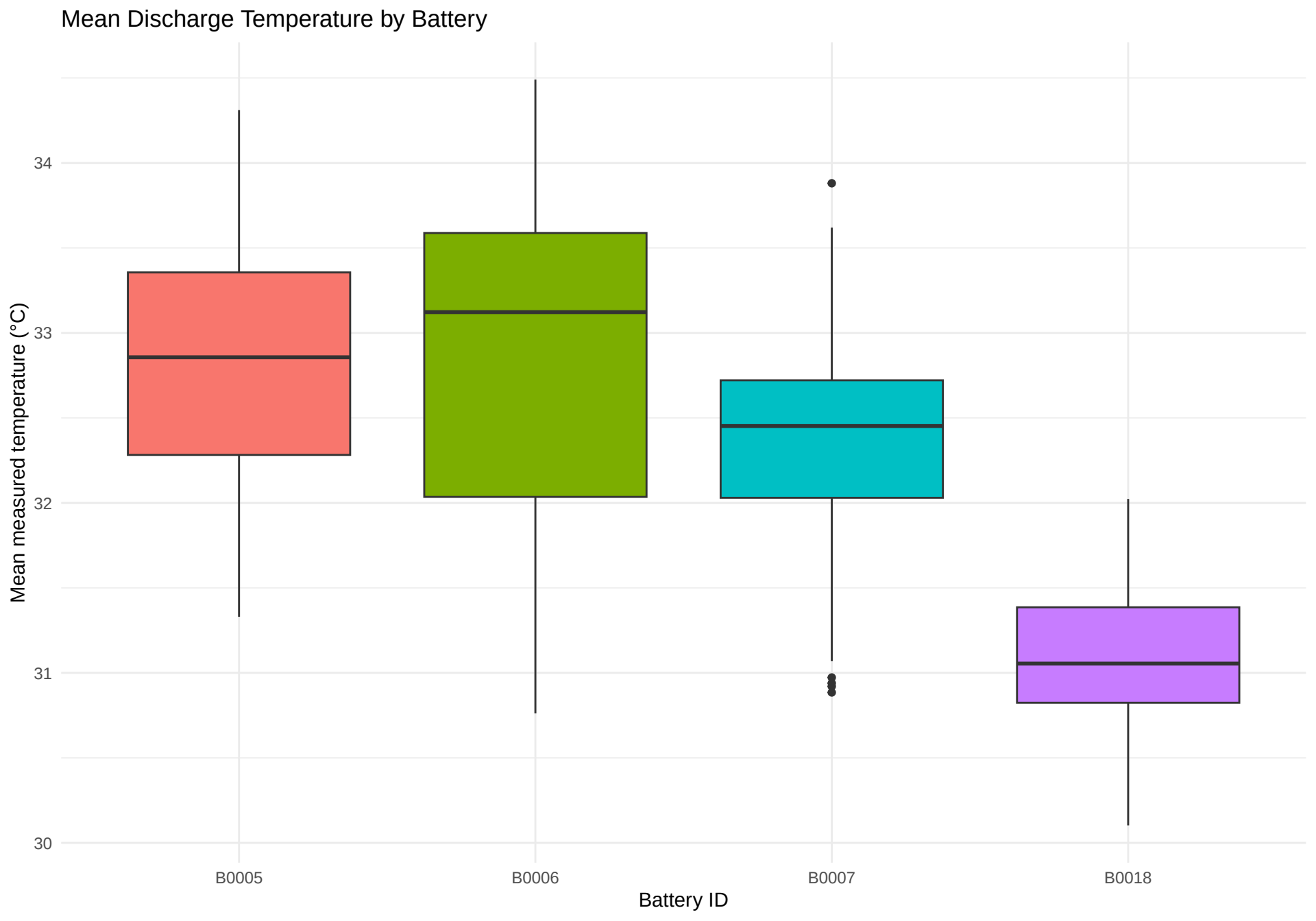

Distribution of Discharge Duration by Battery
Discharge duration (minutes)
60
55
50
45
B0005
B0006
B0007
B0018
Battery ID

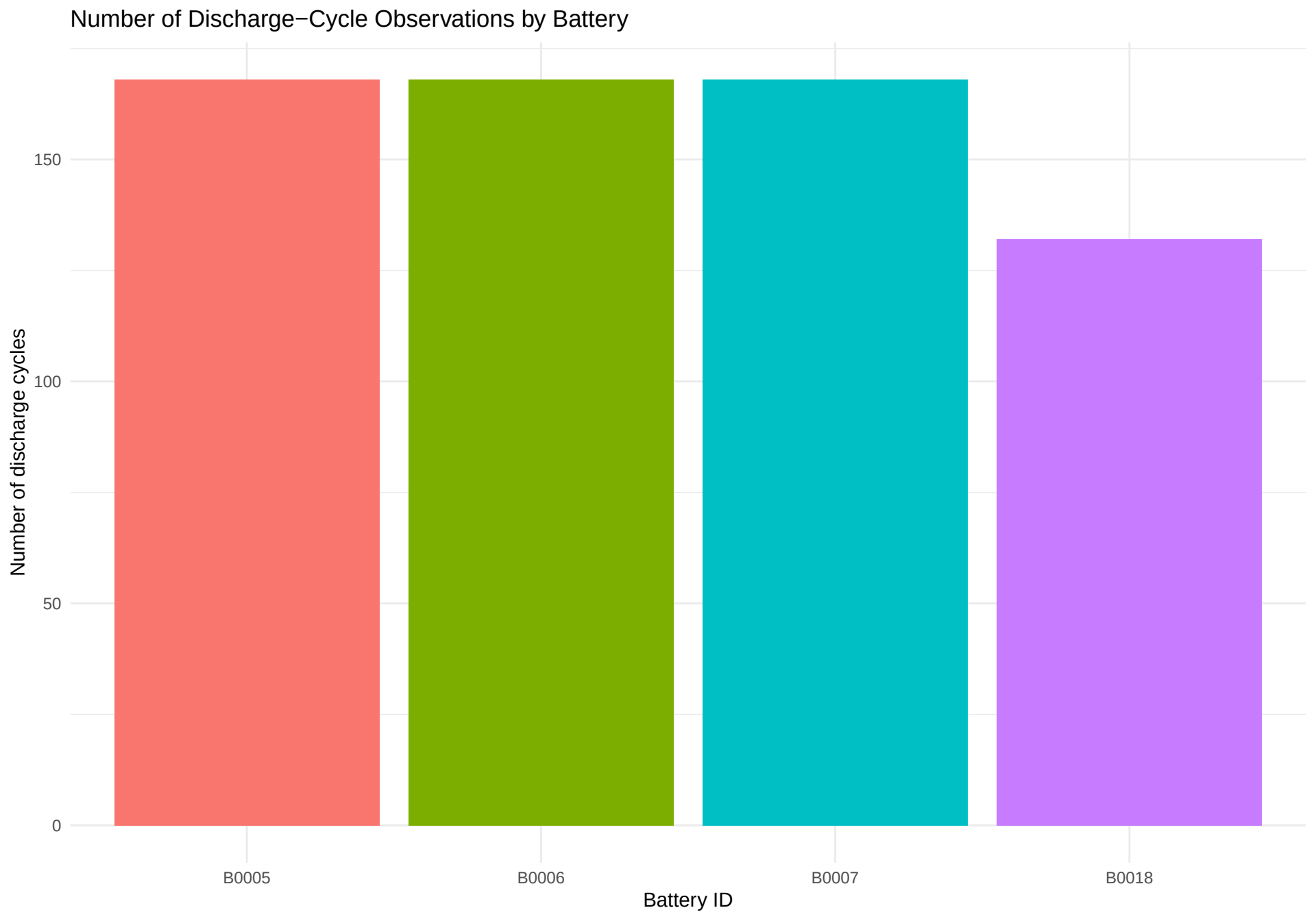

Distribution of Battery Discharge Capacity
Number of discharge cycles
0
20
40
60
1.25
1.50
1.75
2.00
Discharge capacity (Ah)


Battery Capacity Decline Over Discharge Cycles
Dashed line: 1.4 Ah end−of−life reference threshold
Discharge capacity (Ah)
2.0
1.8
1.6
1.4
1.2
0
50
100
150
Discharge−cycle number
Battery ID
B0005
B0006
B0007
B0018


Distribution of Mean Discharge Temperature
Number of discharge cycles
200
150
100
50
0
30
31
32
33
34
Mean measured temperature (°C)


Discharge Capacity by Battery
Discharge capacity (Ah)
2.0
1.8
1.6
1.4
1.2
B0005
B0006
B0007
B0018
Battery ID


Mean Discharge Temperature by Battery
Mean measured temperature (°C)
34
33
32
31
30
B0005
B0006
B0007
B0018
Battery ID


Number of Discharge−Cycle Observations by Battery
Number of discharge cycles
150
100
50
0
B0005
B0006
B0007
B0018
Battery ID

Share of Discharge−Cycle Observations by Battery

